

SOCIOL 723: Problem Set 2

Assigned: Thursday, September 24 • **Due:** Thursday, October 8, 11:59PM

Your name:

Instructions. Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. This problem set is about what you will do in research: reading a published table, fitting models on real data, and writing up what they say. No question asks for a derivation. Every computation uses functions from Labs 3 and 4, and the output printed in Part 1 is read, not re-run. Where a question asks for sentences, write them as you would in a paper. You may use AI assistance, but you are responsible for every line you submit. Use `set.seed(723)` wherever there is randomness.

Data. The GSS extract is `Data/gss_earnings.rds`. The school data are `nlme::MathAchieve` and `nlme::MathAchSchool`, merged as in Lab 4. Part 2 uses one published article, which you obtain through the Duke Libraries.

Part	Topic	Material
1	Maximum likelihood, in practice	Output printed below
2	Reading a published logit	Killewald (2016), <i>ASR</i>
3	Counts	GSS extract
4	Observations in groups	High School and Beyond (<code>nlme</code>)

Part 1: Maximum Likelihood, in Practice

The output below comes from the GSS extract. It fits the logit from lecture, `ba ~ pareduc + wordsum + female + black`, with `control = glm.control(trace = TRUE)`, so that `glm()` prints one line per iteration before the usual `summary()`; `logLik()` of the same model follows.

Two words R uses that the slides did not. Deviance is R's name for -2 times the log-likelihood. With a 0/1 outcome nothing else enters, so the residual deviance is exactly -2ℓ at the estimate, and the null deviance is the same thing for the model with only an intercept; a smaller deviance is a higher likelihood. *Fisher scoring* is R's name for the Newton steps of Weeks 3 and 4 (for a logit the two are the same thing). "Number of Fisher Scoring iterations: 4" means the estimate moved four times, and the trace prints the deviance after each move.

```
                                Output
Deviance = 3971.34 Iterations - 1
Deviance = 3958.874 Iterations - 2
Deviance = 3958.804 Iterations - 3
Deviance = 3958.804 Iterations - 4

Call:
glm(formula = ba ~ pareduc + wordsum + female + black, family = binomial,
    data = gss, control = glm.control(trace = TRUE))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -5.81041    0.24539  -23.678  < 2e-16 ***
pareduc       0.20656    0.01342   15.389  < 2e-16 ***
wordsum       0.39139    0.02404   16.281  < 2e-16 ***
female        0.27863    0.07756    3.592 0.000328 ***
black        -0.11440    0.10982   -1.042 0.297546

      Null deviance: 4751.9  on 3508  degrees of freedom
Residual deviance: 3958.8  on 3504  degrees of freedom
AIC: 3968.8

Number of Fisher Scoring iterations: 4

'log Lik.' -1979.402 (df=5)
```

1. The idea, in one curve.

Let θ be the share of adults with a bachelor's degree, `k <- sum(gss$ba)` successes in `n <- nrow(gss)` draws. For `theta <- seq(0.2, 0.5, by = 0.001)`, compute `dbinom(k, n, theta, log = TRUE)` and plot it against `theta`. This is the log-likelihood: the same `dbinom()` you would use for a probability, read with the data fixed and θ moving. Mark the highest point and confirm it is `mean(gss$ba)`. Two sentences: what does "maximum likelihood estimate" mean, and what does the width of the peak tell you, intuitively, about the uncertainty of the estimate?

2. Reading the likelihood lines.

The model has five parameters.

- (a) Confirm that the residual deviance is -2 times the log-likelihood line.
- (b) Using the AIC formula from Week 3, compute AIC from the log-likelihood line and the five parameters, and check it against R's AIC line.
- (c) The null deviance belongs to the model with only an intercept. What quantity from Week 3 is the difference between the two deviances, and what question does it answer?
- (d) The trace shows the deviance after each of the four steps, and it stops changing after the third. In two sentences, what was the algorithm doing at each step, and why is stopping there right?

Part 2: Reading a Published Logit: Killewald (2016)

Killewald, Alexandra. 2016. “Money, Work, and Marital Stability: Assessing Change in the Gendered Determinants of Divorce.” *American Sociological Review* 81(4): 696–719.

Get the PDF through the Duke Libraries (doi: 10.1177/0003122416655340). Read the abstract, the *Data and Methods* section, and the results around Table 3 and Figure 1 (pp. 708–714); skim the rest.

The question. Two accounts of money and divorce had competed in the literature. In the first, marriage is an economic arrangement: couples with more resources are more stable (*financial strain*), and a wife who could support herself after a divorce is freer to leave (*economic independence*). In the second, marriage is a *gendered institution*: what matters is not the dollars but whether the couple conforms to the expectation that the husband earns and the wife keeps the home. Killewald tests the three ideas side by side and asks whether their weight changed between marriages begun in 1974 or earlier and marriages begun in 1975 or later.

The data. The Panel Study of Income Dynamics has followed the same families since 1968. The analysis file is a stack of *couple-years*: one row for each year a couple is observed married, recording the couple’s circumstances that year and whether they divorced by the following year. A couple contributes a row every year until it divorces or leaves the study. The earlier cohort has 1,908 couples, 381 divorces, and 32,853 couple-years; the later cohort 4,401 couples, 1,303 divorces, and 33,470 couple-years. The outcome in every row is 0 or 1, so the model is a logit.

	Married 1974 or earlier			Married 1975 or later		
	$\hat{\beta}$	(SE)	OR	$\hat{\beta}$	(SE)	OR
Wife employed full-time	.267	(.153)	1.306	.069	(.088)	1.071
Husband employed full-time	−.097	(.208)	.908	−.293**	(.106)	.746
Wife’s proportion of housework*	−1.117*	(.463)	.327	.050	(.233)	1.051
Constant	−7.476***	(.884)		−3.525***	(.393)	
Couples		1,908			4,401	
Divorces		381			1,303	
Couple-years		32,853			33,470	

Killewald (2016), Table 3, excerpt; couples and divorces from Table 2. Logit coefficients, standard errors, and odds ratios as printed. Stars on a coefficient: * $p < .05$, ** $p < .01$, *** $p < .001$, within the cohort. The star on the variable name marks a coefficient that differs significantly between cohorts. Controls include marital duration, religion, race, home ownership, children, premarital cohabitation and birth, age at marriage, both spouses’ education and health, the wife’s occupation and industry, region, and year of marriage.

Reading Table 3. Each cell prints three numbers: the logit coefficient, a change in the log-odds of divorce; its standard error in parentheses; and the odds ratio $e^{\hat{\beta}}$. Stars next to a coefficient test whether it differs from zero within its cohort. A star next to a *variable name* is a different

test: whether the coefficient differs between the two cohorts. The excerpt above keeps the three variables the questions use; the full table has a long list of controls and a third column that pools the cohorts.

Reading Figure 1. A logit coefficient does not tell a reader how much the risk of divorce changes. Figure 1 (p. 713) does. For each cohort it draws the predicted probability of divorce in the next year for a couple with every other variable held at its cohort mean, switching one variable at a time: the husband employed full-time or not, the wife employed full-time or not, the wife doing 50% or 75% of the housework. Because only one variable moves between the two bars of a pair, each pair is one coefficient of Table 3 translated to the probability scale. The table below transcribes its twelve bars.

	Married 1974 or earlier	Married 1975 or later
Husband not employed full-time	1.1%	3.3%
Husband employed full-time	1.0%	2.5%
Wife not employed full-time	1.0%	2.5%
Wife employed full-time	1.3%	2.6%
Wife does 50% of housework	1.5%	2.5%
Wife does 75% of housework	1.1%	2.6%

Killewald (2016), Figure 1, transcribed. Predicted probability of divorce in the next year for a couple with all other variables at the cohort mean; each pair of rows changes one variable.

Questions

The items below build the reading of Table 3 from scratch, one step at a time: the equation, the odds, the odds ratio, the probabilities, and only then the stars. Write every number down; the arithmetic is the point.

1. The data.

- In two sentences, say what one row of Killewald's data is and what its outcome variable records.
- Compute $381/32,853$ and $1,303/33,470$. What is each ratio? Use it to explain why every bar in Figure 1 sits between 1% and 4%, when you may have had lifetime divorce rates of 30% to 50% in mind.

2. The model, written down.

Let p_i be the probability that couple-year i ends in divorce. Killewald's model for the later cohort is

$$\log \frac{p_i}{1 - p_i} = \beta_0 + \beta_1 \text{HusbandFT}_i + \beta_2 \text{WifeFT}_i + \beta_3 \text{Housework}_i + (\text{controls}),$$

where HusbandFT_i and WifeFT_i are 1 if that spouse is employed full-time and 0 otherwise, and Housework_i is the wife's share of the housework, from 0 to 1.

- (a) Rewrite the equation with the four printed numbers from the later-cohort column in place of β_0 to β_3 .
- (b) The left-hand side is the *log-odds*. In one sentence, what range of values can it take, and why does the model put it there rather than p_i itself?
- (c) Read one coefficient from your equation in a sentence of the form “holding everything else fixed, ... changes the log-odds of divorce by ...”.

3. Odds.

The *odds* of an event are $\text{odds}(p) = p/(1 - p)$: the probability it happens divided by the probability it does not.

- (a) Compute the odds for $p = .025, .033, .5$, and $.9$.
- (b) The inverse is $p = \text{odds}/(1 + \text{odds})$; check it on one of your four numbers.

4. From the equation to the odds ratio.

Take two later-cohort couples that are identical in every variable except that the husband is employed full-time in one ($\text{HusbandFT} = 1$) and not in the other ($\text{HusbandFT} = 0$).

- (a) Write the log-odds of each couple from your equation in item 2 and subtract the second from the first. Everything but one term cancels: which term?
- (b) Exponentiate both sides of that difference. The left-hand side becomes the ratio of the two couples' odds, the *odds ratio*, and the right-hand side becomes $e^{\hat{\beta}_1}$. Compute it and confirm it is the printed .746.
- (c) One sentence on each scale: what $-.293$ says about the log-odds, and what .746 says about the odds (as a percentage change).

5. From the odds ratio to probabilities.

This item stays with one variable: the husband's full-time employment in the later cohort, odds ratio .746 from item 4. That odds ratio says the husband's full-time employment multiplies the odds of divorce by .746; it does not say how many percentage points the probability of divorce falls. That depends on the probability the couple starts from.

- (a) Figure 1 gives 3.3% for a typical later-cohort couple whose husband is not employed full-time. Convert 3.3% to odds, multiply by .746, and convert back to a probability. You should land on the bar for a husband employed full-time, 2.5%.
- (b) Now start from an imaginary high-risk couple whose husband is not employed full-time, with a 30% probability, and apply the same .746 for the husband's full-time employment. What probability do you get, and by how many percentage points did it

fall?

- (c) Two sentences: why is the odds ratio the same for both couples while the change in probability is not, and which of the two scales would you give a reader who asks how much a husband's employment matters?

6. Significance, and a comparison across cohorts.

Each coefficient comes with a standard error, and $z = \hat{\beta}/SE$ is the Wald statistic from Week 3.

- (a) Compute z for the husband's full-time employment in the later cohort and say what the two stars mean.
- (b) In the later cohort, the wife's full-time employment has coefficient .069 and standard error .088; Figure 1 shows 2.6% when she is employed full-time and 2.5% when she is not. Compute z and write the sentence that belongs in a results section.
- (c) In the earlier cohort the same coefficient is .267 with standard error .153, and Figure 1 shows 1.3% against 1.0%. Killewald writes that the association declined across cohorts but that the decline is not statistically significant. The two cohorts are separate samples, so the difference between the two coefficients has standard error $\sqrt{.153^2 + .088^2}$. Compute the difference, its standard error, and their ratio, and use them to explain in two sentences what supports each half of her claim.

7. Units matter.

The wife's proportion of housework runs from 0 to 1, and its coefficient in the earlier cohort is -1.117 , printed odds ratio .327.

- (a) The odds ratio .746 for the husband's employment compared two couples whose HusbandFT differed by one unit, 0 against 1. The odds ratio .327 likewise compares two couples whose housework share differs by one unit. On a scale from 0 to 1, what change in housework is that?
- (b) Figure 1 plots a smaller change, from 50% to 75%, so $\Delta = .25$ and the odds are multiplied by $e^{-1.117 \times .25}$. Start from the figure's 1.5% at 50%, convert to odds, multiply, and convert back; you should land on the figure's 1.1%.
- (c) One sentence: why does a coefficient's size mean nothing until you know the units of its variable?

Part 3: Counts

Fit the Poisson regression for the number of children:

```
m_pois <- glm(childs ~ educ + female + black + exper + evermar, data = gss, family = poisson)
```

1. Reading it, and which standard errors.

- Interpret $e^{\hat{\beta}}$ for `educ` and for `evermar`, one sentence each, in the form you would use in a paper.
- Report the standard error of `evermar` from the Poisson fit and from `sandwich::vcovHC(m_pois)`. The Poisson standard error assumes the variance of the count equals its mean; the `vcovHC` standard error does not. Which is larger, and by what percentage?
- Run `AER::dispersiontest(m_pois)`. Two sentences: what does the test say about the Poisson's assumption that the variance equals the mean, and which of the two standard errors in (b) would you report as your default, and why?

2. Zeros, and a model with two parts.

Report the observed share of childless respondents and the share the Poisson predicts (average `dpois(0, fitted(m_pois))`).

Fit the hurdle model with `pscl::hurdle(..., dist = "poisson")` and the same covariates in both parts. Make one table with the log-likelihood, number of parameters, AIC, and BIC of the Poisson and the hurdle.

Then read the `evermar` coefficients from the hurdle model's two parts. In a paragraph written for a reader, say what `evermar` does to the chance of *any* child (the zero part), what it does to the number of children *among parents* (the count part), and what the single Poisson coefficient was averaging over.

Part 4: Observations in Groups

Merge `nlme::MathAchieve` with `nlme::MathAchSchool` as in Lab 4, and convert `School` to a plain factor with `factor(as.character(School))`, as the lab does: 7,185 students in 160 schools, math achievement, student SES, and school sector.

1. From clustered standard errors to a random intercept.

Fit `lm(MathAch ~ SES + Sector)` and report the coefficient on `Sector` with its plain standard error and with its standard error clustered on `School`.

In a short paragraph, explain to a first-year student why the plain standard error is wrong for this variable: sector is a property of the school, so it varies across 160 schools, not across 7,185 students.

Then fit `lmer(MathAch ~ SES + Sector + (1 | School), REML = FALSE)` and report the two coefficients with standard errors, $\hat{\tau}^2$, and $\hat{\sigma}^2$.

Say in one sentence each what the two variances measure, compute the intraclass correlation, and close with two sentences on what the random-intercept model gives you that the clustered standard error did not.